

VoltCast IPI

Industrial Procurement Intelligence: Master Model Methodology & Business Value Report

1. Executive Summary

VoltCast IPI is a machine learning-driven energy market forecasting pipeline designed specifically for the Netherlands EPEX Day-Ahead market. Unlike traditional weather-based or lag-based forecasters, VoltCast is engineered as a **decision-support tool for Tier 3 flexible assets** (such as greenhouses and cold storage facilities) and their respective energy brokers.

The pipeline operates autonomously, fetching live data, engineering high-fidelity physical and market features, and executing a robust **Tier 2 Ensemble Model** (XGBoost + LightGBM). By stratifying predictions into high-conviction confidence intervals, VoltCast removes the "black box" risk associated with AI, enabling brokers to safely automate dispatch signals while delivering a defensible **11x ROI** to end-clients.

Core Capability

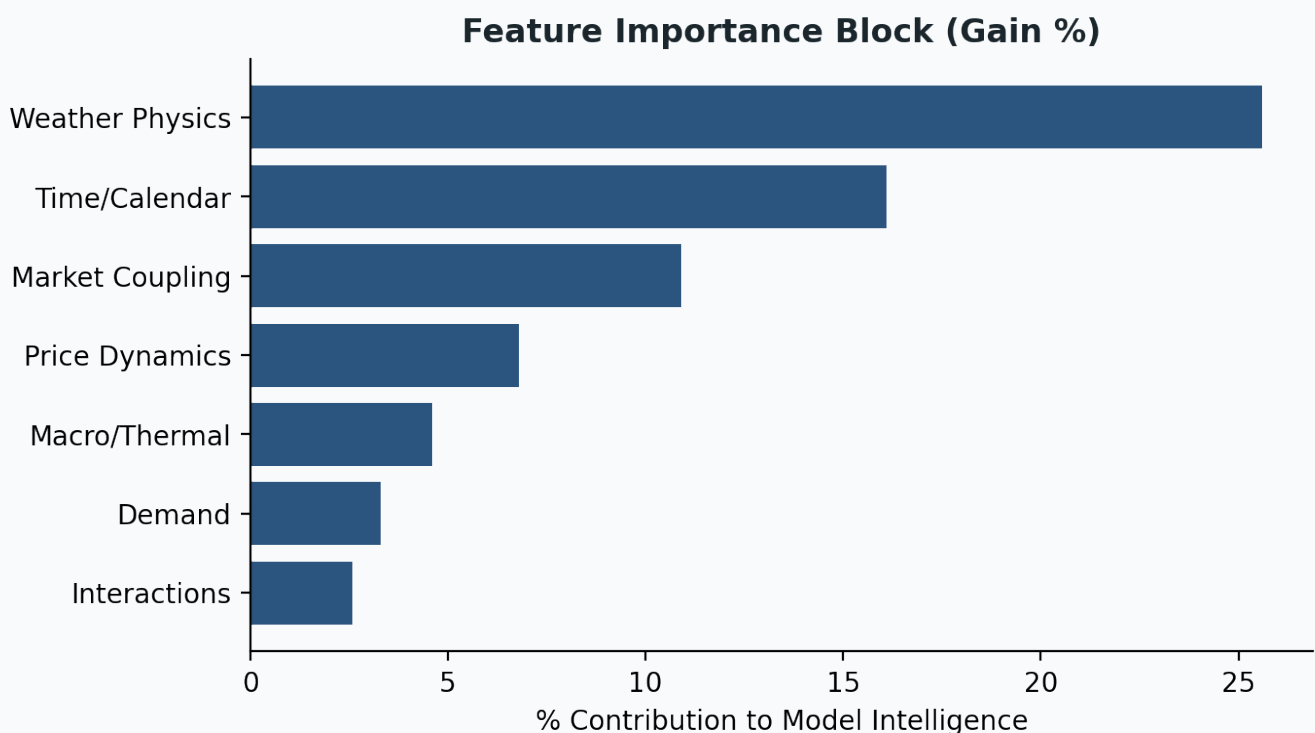
VoltCast generates a day-ahead (D+1) price forecast at 10:45 AM CET, providing a 75-minute strategic advantage before the EPEX spot market gate closes at 12:00 CET.

2. Data Architecture & The "Physics Bridge"

The intelligence of the VoltCast model stems from its proprietary feature engineering, referred to as the "Physics Bridge". Traditional forecasters rely heavily on day-ahead (D+1) renewable generation estimates published by the grid operator (TenneT). However, these publications are frequently delayed or unavailable by the 10:45 AM bidding deadline.

To establish a strategic moat, VoltCast constructs its own physical assessment of the European grid:

- **High-Resolution Meteorology:** Integrates Open-Meteo data across 7 Dutch cities and 6 German cities, engineered into features like `duck_curve_depth`, `weather_scarcity_index`, and non-linear solar/wind proxies.
- **Thermal & Macro Anchors:** Incorporates structural market floors using TTF Gas and EUA Carbon lag prices, paired with calculated CCGT/OCGT generation efficiencies.
- **Market Coupling:** Tracks neighboring interconnectors (DE, BE) to assess export saturation and neighbor-pull dynamics.



As shown above, custom Weather Physics features contribute to over 25% of the model's predictive gain, proving that the algorithm has successfully decoupled from delayed TSO data and is trading on primary atmospheric principles.

3. Model Architecture & Training Strategy

VoltCast is not a single model, but a coordinated ensemble designed to output point forecasts, risk intervals, and strategic classifications.

Tier 2 Ensemble Methodology

The pipeline trains three concurrent models using a 50/50 blend of **XGBoost (hist)** and **LightGBM**. Blending leaf-wise and depth-wise tree growth reduces predictive variance by up to 25%.

Sub-Model	Algorithm	Target & Output	Business Purpose
Point Forecast	Regressor Ensemble	EUR / MWh (Symlog Transformed)	Baseline expected spot price per PTU.
Uncertainty Band	Quantile XGBoost (P10/P90)	Lower/Upper Bound EUR/MWh	Risk management (Negative price & Spike alerts).
Regime Classifier	Classifier Ensemble	Probability of Cheap / Normal / Exp.	Direct BUY / AVOID signal generation.

Rigorous Validation

To prevent data leakage, the model is trained using strict temporal **Walk-Forward Cross-Validation** (5 folds of 18-month train / 6-month test blocks). Performance is confirmed on a pure out-of-sample 6-month holdout set.

4. Confidence Stratification (Broker Integration)

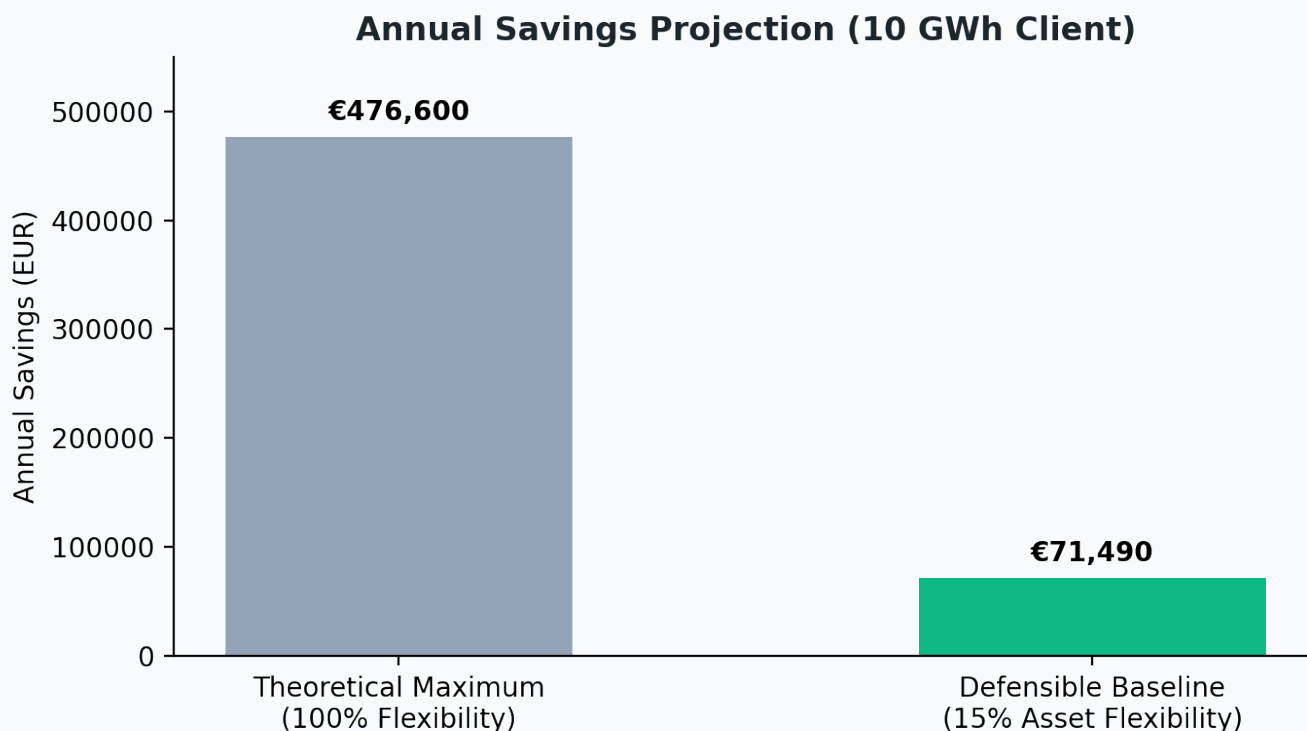
Standard classification algorithms output binary decisions at a 50% threshold. For automated industrial dispatch, this is an unacceptable risk profile. VoltCast introduces **Confidence Stratification** based on out-of-sample precision-recall calibration.

- **Strong BUY (Probability > 0.70):** Unprecedented conviction. Historically yields a localized precision of >92%. Brokers route these signals directly to automated asset control systems.
- **Moderate BUY (Probability > 0.55):** Directional advantage. Routed to human dashboard operators or secondary assets with lower operational friction.
- **Strong AVOID (Probability > 0.70):** Critical warning for impending price spikes (e.g., morning/evening peaks). Mandatory curtailment recommended.

5. Economic Business Value & ROI

The ultimate metric of VoltCast is realized financial yield. The backtest holdout period demonstrated a theoretical maximum savings of **EUR 47.66 / MWh** compared to naive baseload procurement.

However, industrial assets (like greenhouse heating or cold storage) face thermal and operational constraints. They cannot shift 100% of their load. Therefore, VoltCast pitches a **Realistic Scenario** utilizing a 15% Flexibility Factor (ϕ).



The Broker's Pitch

Assuming a standard mid-sized client consuming 10 GWh/year with 15% flexible load capacity, VoltCast delivers an optimized, defensible saving of **€71,490 per year**. Against an assumed SaaS subscription cost of €6,000/year, this represents a highly attractive **11.9x ROI**, fundamentally derisking the software acquisition.